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Additive Manufacturing Cyber-Physical System: Supply Chain Cybersecurity and Risks

NIKHIL GUPTA[1], AKASH TIWARI[2], SATISH T. S. BUKKAPATNAM[2], AND RAMESH KARRI[1], (Fellow, IEEE)

[1]Center for Cybersecurity, New York University Tandon School of Engineering, Brooklyn, NY 11201, USA
[2]TEES Institute for Manufacturing Systems and Department of Industrial and Systems Engineering, Texas A&M University, College Station, TX 77843, USA

Corresponding author: Nikhil Gupta (ngupta@nyu.edu)

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ABSTRACT Additive Manufacturing (AM) methods have become increasingly efficient and industrially viable in the past ten years. These methods offer the freedom of complexity to the designers and choices of localized and pull-based production system to the managers. These propositions of AM have been enabling custom manufacturing and are catalysts for rapid growth of additive manufacturing (AM). This paper analyzes the general characteristics of AM supply chain and proposes three AM supply chain models based on the specific nature of the industry. Our description of the models emphasizes on adopting an holistic view of the AM supply chain and therefore includes raw material, printer hardware and the virtual supply chain. Throughout the product life cycle of additively manufactured products, the interlacing of the virtual supply chain (*digital thread*) with the physical supply chain and their operations fundamentally make the AM process a cyber-physical system (CPS). Therefore, the technology brings along with it benefits of a CPS as well as a new class of attack vectors. We discuss the possible attacks (printer, raw material and design level), risks (reverse engineering, counterfeiting and theft) and provide an enhanced risk classification scheme. We contend that the traditional cybersecurity methods need to evolve to address the new class of attack vectors that threaten the AM supply chain and also discuss the nature of existing solutions that help in addressing the risks and attack threats. In providing an holistic view of the AM supply chain the interdependencies of the processes in the AM supply chain are presented and we elucidate the effects of local attack vectors on the entire supply chain. Further, we discuss the existing security measures to mitigate the risk and identify the existing gap in AM security that needs to be bridged.

INDEX TERMS Manufacturing technology, supply chain management, risk analysis, production engineering.

I. INTRODUCTION

Additive manufacturing (AM) offers two unparalleled value propositions – customizability while maintaining potential profitability and freedom in design complexity. Industries have leveraged the proposition of this technology for catering to their niche requirements. The pharmaceutical manufacturing industry is capable of producing personalized drug dosage based on patient's mass and metabolism [1]. Releasing the constraint of complexity in design is leveraged by the biotechnology industry to provide patients with anatomically specific prosthetics including bone implants [2], heart valves [3] and tracheal splints [4]. The automotive, aerospace and electronics industries have also leveraged upon both propositions in their production processes. With advancements in materials science, the users can enjoy the benefits of *custom manufacturing*, where the functionality of the product is not compromised due to the limitations of the fabrication method [5].

Implications of these propositions of AM technology to a manufacturer's supply chain are profound. Implementation of AM technology in manufacturing allows for realization of lean, agile and sustainable supply chain. AM is bringing manufacturing closer to the consumers, therefore avoiding the physical part from having to go through multiple-tiers of traditional supply chain, more so that manufacturing using polymer based materials can be done at the end use location. Therefore, AM technologies have been attracting participants

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of the maker movement who help in realizing open source manufacturing. RepRap is one such open source project that has successfully created a 3D printer which besides other products, can print its own parts and is thus a self-replicating manufacturing machine [6].

AM offers flexibility to choose where the manufacturing occurs in the supply chain. This reduces the constraints involved in planning and laying out the supply chain; offers benefits of sustainability and profitability. Strategically located AM hubs help improve the utilization of traditional manufacturers with excess capacity by making them enablers of the hybrid-AM practice [7]. AM can obviate many of the problems that exist in the traditional manufacturing supply chain such as inventory buffering of a large number of components, inventory of tools, and multi-tier supply chains [8]. It reduces the logistics involved by replacing physical delivery with digital instructions that can be used by the end users to manufacture the products at the location of use [9]. Last mile problems and out-of-stock problems can be addressed by 3D printing at the point of use itself. The proposition of a localized and a pull-driven production enables on-site and on-demand manufacturing. This contributes towards the improved agility of supply chain with respect to increased service levels while reducing delivery time, aggregate inventory levels, inventory risk and pipeline stock [10].

The novel supply chain models that emerge from the implementation of AM technology become conspicuous – drug manufacturers, following a distributed manufacturing paradigm, can outsource manufacturing to the point of sale; end-users manufacture designs at home or at cyber-manufacturing kiosks [5]; manufacturers of automotive and aerospace parts may continue to maintain a multi-tier supply chain, while leveraging the benefits of being able to realize complex parts for their engineering applications.

These valuable benefits of implementing AM in the supply chain takes root in the maintenance of a *digital thread* and its digital design process. A digital thread for manufacturing refers to the data and information flow generated through a product life cycle [11]. It captures the salient events as well as states of the manufacturing system starting from the initial conception and design of a product through its modeling, production, monitoring and use. One of the main research thrusts of the manufacturing community for the past 30 years has been to harness the digital thread to create a computational framework—nowadays known as a *digital twin*—to emulate and analyze the product and the process before (for design and process planning), during (for quality assurance) and after (diagnostics and maintenance) the realization of a product [12], [13]. The microelectronics industry have embraced the digital thread for computer-aided design (CAD) of the electronic components and devices and digital testing [14]. Opportunities for extending digital thread-driven design (based on composing the functional elements) and life cycle management practice beyond electronic systems became possible with layer-wise material deposition in AM for a wide variety of geometries and application domains.

A combination of digital design process (initial stage of the product life cycle) with physical manufacturing path makes AM a cyber-physical system (CPS). With AM, most of the design processes and its iterations can become collaborative and take place over the internet. The designer delivers the digital files, which carry the design of the parts, online. Although the physical supply chain is vastly simplified, the supply chain in AM takes a hybrid physical - digital form [7].

Much of the capabilities of AM enjoyed by manufacturers, supply chain managers and end users come with a call for attention to develop a robust security infrastructure around their use of this technology. The integration of digital thread with the physical supply chain increases the attack surface and makes the AM supply chain vulnerable to cyber-attacks that can lead to defects in physical products. The vulnerability of this technology and its supply chain poses security risks to printing and distributing critical parts for engines, defense equipment and pharmaceutical drugs and devices. Although the trail of physical supply chain of the printed part may be reduced, the end user may still need to assume four functions– procurement of raw materials; the AM design blueprints and digital printing instructions; AM hardware such as an appropriate 3D printer and quality assessment of the manufactured part. Each of these activities are still susceptible to threats. The network of stakeholders, involved from the raw material sourcing to the final product manufacture, constituting the supply chain may still be susceptible to compromise and counterfeiting [9]. There are fewer and more obvious critical raw material suppliers of plastics, metal alloys, ceramics and other emerging feed materials but their number is increasing as the technology is advancing [8]. Additional cyber risks may include information breaches pertaining to location of supply in transit, associated financial transactions and contracts, intellectual property (IP) and digital usage rights. However, these risks are not unique to AM field and may not directly compromise the product quality. Most-cyber-physical systems are exposed to such risks.

The early inception of the semiconductor industry makes it more mature than the emerging AM-based custom manufacturing industry, but is still trying to cope with vulnerabilities of the digital thread like the latter. CAD enables the designers with the processes of design and logic synthesis, verification and analysis, circuit layout and optimization, post fabrication validation and functional testing [14]. Analogous process can also be found in the digital thread of AM.

The nature of risks in the AM supply chain is such that sabotage of either the manufacturing process or supply chain obviates any value created through the process chain. Any tangible solution requires simultaneous consideration of both these chains. There continues to exist a gap between the risk mitigation solutions for process chain and their implementation within the supply chain.

Supply chain risk management (SCRM) frameworks involve stages of risk identification, risk assessment and risk mitigation [15]. In addition, due to the cyber-physical nature of AM, cyber risk management of the AM supply chain

is also required to be addressed. Independent developments have been observed across the aforementioned stages for the AM supply chain. Thorough survey on risk taxonomy for additive manufacturing is presented in previous published works [16], [17]. Developments have taken place in mitigating risks of counterfeiting [18], [19], transparency [20], [21] and the manufacturing process [22], [23] in the AM security. Nonetheless, the extant body of literature of SCRM for AM remains preliminary and conceptual [24]. There exist no studies that place these developments in context of mitigating the AM supply chain risks. Further, development of holistic methods to capture risk factor interdependencies across the entire supply chain network and propagation of disruptions in the supply network remain as important research gaps in the literature of SCRM [25].

We propose a unified view of observing the AM manufacturing process chain and its supply chain as a CPS. This unification allows us to assess the implications of the various risks – process chain and supply chain – with the introduction of the AM technology. The unified view serves as a guide in identifying and implementing existing solutions, and reveal security gaps that requires attention.

In this study, the manufacturing process chain and the supply chain have distinct connotations. The process chain spells out the process that a CAD file undergoes from model development to printing a physical part on a 3D printer. It shows the technical process flow for this file. In comparison, the supply chain means the entire chain of events from the production of raw materials to the delivery of the final product to the end use customer. The process chain is a sub-component of a supply chain. We establish these definitions to streamline this study describing risks and threats at the different stages of the supply chain. This explanation of a process chain dispels the common misconception that the CAD file is a direct input to a 3D printer, while there are transitional steps between generating a CAD file and manufacturing the part on a 3D printer.

In Section II that follows, we explain the manufacturing process chain of AM in detail, from design to print. In section III we present discussion on the three supply chain models that have emerged in industry as a result of AM; each model is supported by a case study. Here we also provide a holistic view of the entire AM supply chain. In section IV we introduce the security issues that exist throughout the AM supply chain and discuss existing solutions and gaps. Finally, in section V we summarize the contributions of the paper.

II. AM PROCESS CHAIN

Fig. 1 illustrates various steps involved in the AM process chain, including developing a CAD solid model and the follow up steps needed for preparing the files for 3D printing [26]. A solid modeling software is used to develop a 3D model of the product. The design phase can be protracted and may require several iterations. An array of analysis software is available for finite element analysis (FEA), computational fluid dynamic analysis and optimization. They are used to

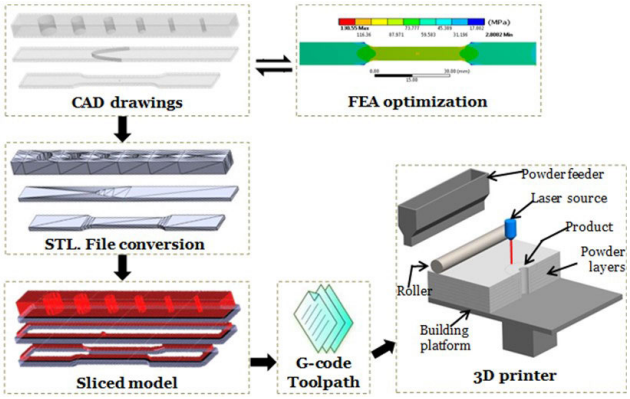

FIGURE 1. The process chain model for AM demonstrating flow of CAD model from development till component printing.

develop and "virtually" test the CAD models. These results can guide design modifications until they yield results that meet the standards set for the properties and performance of a component. For example, design of a partition wall used between aircraft cabins requires conducting bending test, impact test, and weight optimization, all of which can be performed virtually using software. Since a variety of CAD software are available, they convert the developed models from the proprietary format to a generic STL (stereolithography) format. Further steps to process this file include slicing the design in two-dimensional slices that will be printed and then generating the tool path that the print head (extrusion nozzle or laser source) will follow. These steps can be conducted in open source programs or proprietary software programs that may be specific to the printer to capture printer specific information such as slice thickness or diameter of the extruded filament. The tool path file also contains information on printer settings such as temperature or laser power and is used by the 3D printer to print the component. After printing the parts, additional post-processing steps such as support material removal, sectioning the part from the build plate, surface finishing, and heat treatment may be conducted as necessary. In addition, testing of the part using destructive or non-destructive methods is also routinely conducted to find defects and properties.

From the point-of-view of supply chain, the steps of process chain can all be conducted in-house or can be distributed to various parties involved in the supply chain. However, discussion in the following sections will not specifically focus on individual steps of process chain but will take a holistic view from the perspective of various manufacturing and application scenarios of additively manufactured parts.

III. AM SUPPLY CHAIN MODELS

Several supply chain models are available depending on the industry sector. A typical traditional supply chain relevant to industry sector that is engaged in high volume manufacturing is presented in Fig. 2. Much of the material flow in this supply chain follows a linear movement to execute a process chain consisting of a primary process, such as casting,

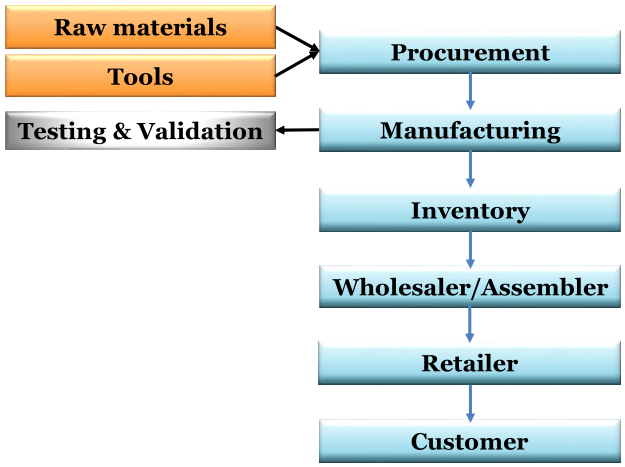

FIGURE 2. A typical supply chain model followed by a large-scale manufacturing sector engaged in traditional manufacturing.

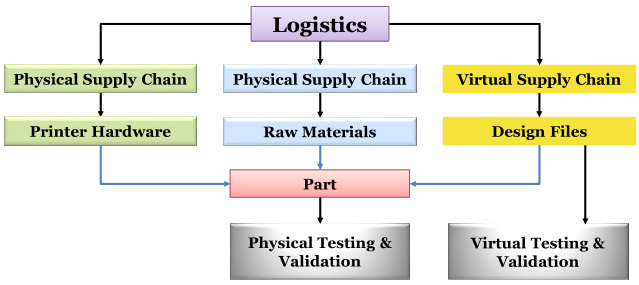

FIGURE 3. The supply chain model for AM.

followed by forming, machining, heat treatment, grinding, or finishing. The industry acquires raw materials and tools to manufacture the component based on the designs that do not change frequently. The manufactured components pass through a business-to-business chain for assembly, packaging, distribution and sale to the final customer.

However, because of the existence of a digital thread, the part now resides in a digital form and can be converted to its physical form on demand where required. This resolves the constraint of the linear flow (Fig. 2) of parts in the supply chain. We envision a general description of the model of the AM supply chain in Fig. 3. Based on logistics, we break up the supply chain into three parallel streams. If the users manufacture the part on-site, they can deploy it out of the print bed. Otherwise, the part can pass through a supply chain that will resemble the traditional supply chain and will lead to the distributor and the user.

Two of the three streams named in Fig. 3, including 3D printer and the raw material used for printing, are hardware movement streams. The third stream includes the design files for the CAD model, which is a virtual stream that may cover software providers, cloud storage provider, CAD model developer, virtual test analysts, and professionals that prepare the final files for 3D printing as per the AM process chain presented in Fig. 1.

Manufacturers using the classical methods for large production runs have a substantial cost and time associated with producing dies, punches, and machining tools. In those methods designers use the CAD files to visualize the component design, while the eventual quality of the component involves the skill level of the operator, e.g. a foundryman or a machinist. In comparison, the printing instructions generated from the CAD files serve as the input in 3D printers and the quality of the part is independent of the skills of the operator. Here, the intrinsic significance of the CAD files is substantial in AM supply chain. In addition, designers can customize the product design, even make it one-of-a-kind, for AM produced parts. This establishes the CAD models to be a valuable IP. Given the suite of software and computer hardware needed for producing, storing and transporting high-value CAD files, Fig. 3 has formed a corresponding logistics stream. The supply chain presented in Fig. 3 can be adapted for various products and industrial scenarios. Three models are discussed in the following sections.

A. MODEL 1

The model presented in Fig. 4 is based on the large-scale production runs scenario resembling conventional manufacturing. Here, the product is not a customized high-value part but requires AM for a special functionality such as jointless inter-connected rings or heat exchangers with an internal channel design. The manufacturer owns the 3D printer and buys raw materials for printing. They deliver the printed parts to the assembler or the parent company that owns the design IP. One generally accepts these original equipment manufacturers (OEMs) to be trustworthy.

Case study: Automotive industry is one example of such a supply chain. Many Tier-1 suppliers manufacture and deliver one or a few components each to the auto company. The auto company assembles the parts and delivers the cars to the dealers for sale. The auto company provides the CAD models of the components to the manufacturers along with the instructions such as dimensional tolerances, surface finish and heat treatments. The manufacturer cannot adjust the design or specifications because each component is part of an assembly, but the burden of quality and performance of the part lies with them. In case of the product recall, the manufacturer may have to take the entire burden of replacing the part.

B. MODEL 2

In this model, the actual product of a company is a design file, which is provided to the customer. The customers may print

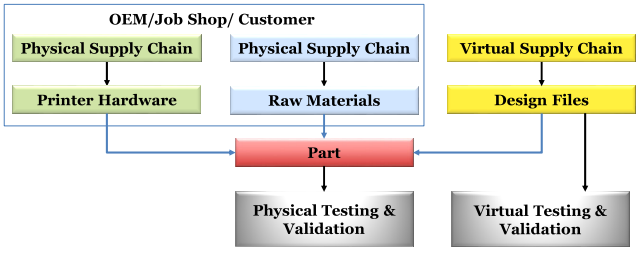

FIGURE 4. Model 1 of AM supply chain that can be applied to examples such as an automotive component.

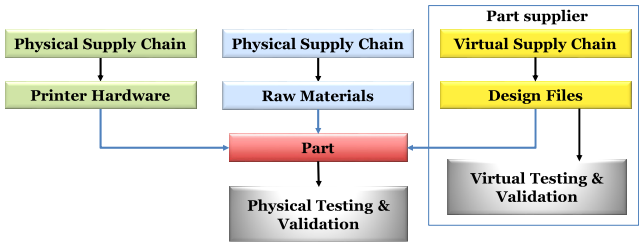

FIGURE 5. Model 2 of AM supply chain where the design file is the actual digital product that is supplied to the user.

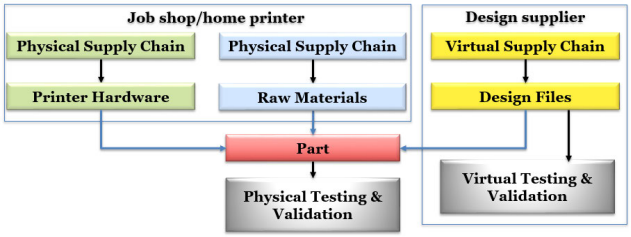

FIGURE 6. Model 3 of the AM supply chain that can be applied to scenarios such as classic cars or aftermarket parts.

the files themselves or have it printed by a job shop for their application. The scenario for this model is presented in Fig. 5.

Case study: One possible example of this kind of supply chain is a part design developed by a contractor for NASA to deploy on International Space Station (ISS). ISS now houses a 3D printer and is capable of printing replacement parts. In this scenario, a contractor may develop a part design, which will be printed and used by astronauts at ISS with no control over the printing or application conditions by the contractor. In this case, mitigation or minimization of risk is a relatively complex scenario. First the contractor may conduct extensive virtual testing and validation on the developed part design. This design may be printed using the same type of printer as the one installed on ISS and tested. A large number of parts would be produced to measure not only the average properties but also to document the standard deviation, which will also provide the worst-case scenario for the properties of that component due to possible printing defects. After developing sufficient confidence in the quality of the printed part, the design will be printed at ISS by astronauts, will be inspected and tested using available methods. A combination of these limited test results in space and on the ground based extensive test results will be used to take the decision about deployment of the part. The designer and the user may share the responsibility if the part does not perform as per the expected standards.

Another example is the case where drug manufacturing is out-sourced to the point-of-sale pharmacy when there is a stock out condition. Drugs are 3D printed as per the design and 3D printing instruction of the pharmaceutical companies like GlaxoSmithKline as reported in some news articles [27]. In this case responsibility of quality of the drug lies on both on the company and the point-of-sale pharmacies. If the drug does not perform its intended function, the 3D printing instructions will be investigated for correctness and the processes of the point-of-sale pharmacy that printed the drugs will be investigated, to check if the instructions to print the drug were followed correctly.

In these scenarios, the CAD model files are the main IP and valuable objects because no physical part printed on the earth will be transported for actual deployment in the case of ISS. Nor can the delivery of medical drugs be expedited beyond a practical scope of time in case of unexpected demand surge resulting in stock-out or a patient requires treatment in a place that is difficult to reach. In both cases, assurance of quality of the actual product that is deployed remains a major challenge. Printers that have embedded capabilities of testing and quality evaluation are the only possible solution.

C. MODEL 3

In this model, presented in Fig. 6, the physical supply chain, including printer and raw material, is under the control of the same entity such as a contract-based print shop, whereas all the design files and the virtual processes are separated in a unified entity. This model is different from Model 2 based on the value of the design files. The focus of the Model 2 is the uniqueness of IP embedded in the design files resulting in the collaboration between designer and part producer, whereas this model has a heavy focus on manufacturing quality and the IP may or may not have a very high value but comes from a different source such as online file repositories.

Case study: The classic car market has a huge demand for replacement parts for many out of production car models. OEM parts for many of the models are not available. Secondary manufacturers step in to fill this demand. However, making cast or forged parts involves maintaining an inventory of dies and tools for each part, which is costly. This limits their capacity of producing all parts based on their capacity for investment in making and storing the tools. AM is filling this niche by enabling manufacturing without using shaping tools. In this case, the customers are themselves responsible for finding the right design files (or may purchase from a marketplace) and then decide to either print the part on their own printer or in a job shop. In many scenarios in this case, the liability of failure may be on the user themselves if they develop their own designs or print the part using their own printer.

AM job shops are enabling a broad range of actors to produce and supply these parts. Skilled designers may develop CAD models from scratch or download them from online repositories and print them in a 3D printing job shop. In these cases, the burden of ensuring the product quality is on this actor that is taking the CAD models from one source and printing the part at a job shop.

D. EXTENDED AM SUPPLY CHAIN

In this section we provide granular details of the three parallel supply chain streams discussed in the previous section. The addition of the enterprise supply chain further extends the

previous models and provides a holistic view of the entire AM supply chain model through unification of the former with the process chain, physical supply chain and the virtual supply chain. This extended supply chain model is illustrated in Fig. 7. In this figure, we identify each component (actor, process, resource and output) of the supply chain by a label. The labels start with a letter followed by a number. The numbers are ordered in the direction of the flow of the supply chain. The letter labels are P (Printer supply chain), R (Raw material supply chain), D (Digital Files supply chain) and S (Enterprise supply chain). The holistic view allows us to observe the interdependency of the processes across the entire AM supply chain.

In the Printer hardware supply chain (P), the manufacturer of AM machines (E.g.: OPTOMEC, ExOne) – the AM OEM (P2) – procures various parts and sub-assemblies from its multiple AM Part Vendors (P1). P2 then produces the AM machine by assembling together the hardware and software (P3.1, P3.2, P3.3 and P3.4). The assembled AM machine (P4) is the final output from the printer hardware supply chain and is ready to be delivered to its customer in the Enterprise supply chain (S). After the AM machine is procured by the Enterprise supply chain, there continues to be maintenance support (P5) from the printer hardware supply chain.

Next, in the Raw materials supply chain (R), the manufacturer of feedstock (R2) procures the primary raw materials from its supplier's distribution network (R1). Production (R2) is followed by Packaging (R3), storage (R4) and Distribution (R5). The final output from this supply chain is the feedstock, reaching into the Enterprise supply chain (S) from the distribution network (R5).

In the Design Files Virtual Supply Chain (D), the chain starts with experts collaborating over a network (D2) to design the CAD model (D2.1), test the model (D2.2) and ensure the fitness for 3D printing (D2.3). The final output from this supply chain into the Enterprise supply chain (S) is the 3D printing instruction (D3) that will be a direct input in the AM machine (P4). The series of steps between D2 and D3 have already been illustrated in detail in fig.1. Alternatively, the 3D printing instructions (D3) may be downloaded from an online marketplace (D1).

Finally, the Enterprise supply chain (S) is representative of a generic manufacturing supply chain. The tier 1 supplier (S1) provides parts and sub-assemblies to the OEM (S2). S2 may also manufacture some of the other parts in-house, followed by assembly (S3), packaging (S4) and delivering the final product it to its customers – the end-user (S6) through its Multi-tier distribution network (S5). Within the Enterprise supply chain, we have included a cyber-manufacturing kiosk (S7) [5], which is equivalent to a job shop, catering to the personalized manufacturing needs of the end user and may be equipped with advanced capabilities of the likes of a hybrid manufacturing system along with a digital interface. S7 is not involved in the linear flow of S1 through S6.

In the previous three models and case studies, we observed that when the responsibilities of the three parallel supply chain streams are combined uniquely for the different stakeholders, we observe new supply chain models based on the use cases, each with their nuances. The responsibilities for quality and liability rests at different places based on these models. Alternative supply chain models may exist or new models may emerge depending on the Enterprise model, value of the part, and location of the user.

Supply chain model 1 is realized when the OEM (S2) procures D3 and delegates part production and responsibility of the quality to S1. S1 procures both P4 and the feed stock from R5. Supply chain model 2 is realized when the end user (S6) procures D3, P4 and feed stock from R5 and prints at location of use itself. Finally, supply chain model 3 is realized when S6 procures only D3 and delegates the printing operation to S7, in which case, S7 procures P4 and feed stock from R5. It is clearly observed how supply chain models 2 and 3 obviates the requirement of S1 through S5 in the Enterprise supply chain. Businesses therefore have the choice of shortening the physical supply chain by manufacturing the part closer to the consumer, or allowing the consumer to manufacture his own parts, by digitally distributing the design and printing instructions for parts.

The intermediate processes involved in the AM supply chain are complex with distributed responsibilities. The supply chains are interlaced and no part of these supply chains can be overlooked if we want to secure the supply chain. In this section we combined and illustrated the different supply chains that exist in the current practice. However, to study the security aspects we need to identify vulnerabilities in each stage of the supply chain. A study of the flow of the physical (raw material, printed parts) and virtual (design files) goods and the risks relevant in these scenarios can help in advising and forming supply chain models for the specific part. In the following section, we will highlight the security vulnerabilities and risks in processes, infrastructure, and the actors that exist throughout the AM supply chain. The virtual supply chain itself is a source of vulnerabilities and its resilience is only as good as the cyber-security infrastructure that it employs.

IV. RISKS AND SECURITY METHODS IN AM SUPPLY CHAIN

A. SUPPLY CHAIN ATTACK CLASSIFICATION

The physical and virtual supply chains are subjected to different risks. In addition, the difference in the fixed installation such as printer and the consumable such as the feedstock are subjected to different risks and attack vectors. Studies are available that have presented threat taxonomy in AM [16], [28]. However, much of the available description is presented from the perspective of cybersecurity and do not address the physical part of the AM CPS. The threats and risks go well beyond traditional cybersecurity risks, such as Denial of Service, hacking to sabotage and ransomware attack, in the field of AM and a considerable enhancement in the discussion is required to cover additional possibilities related to the quality

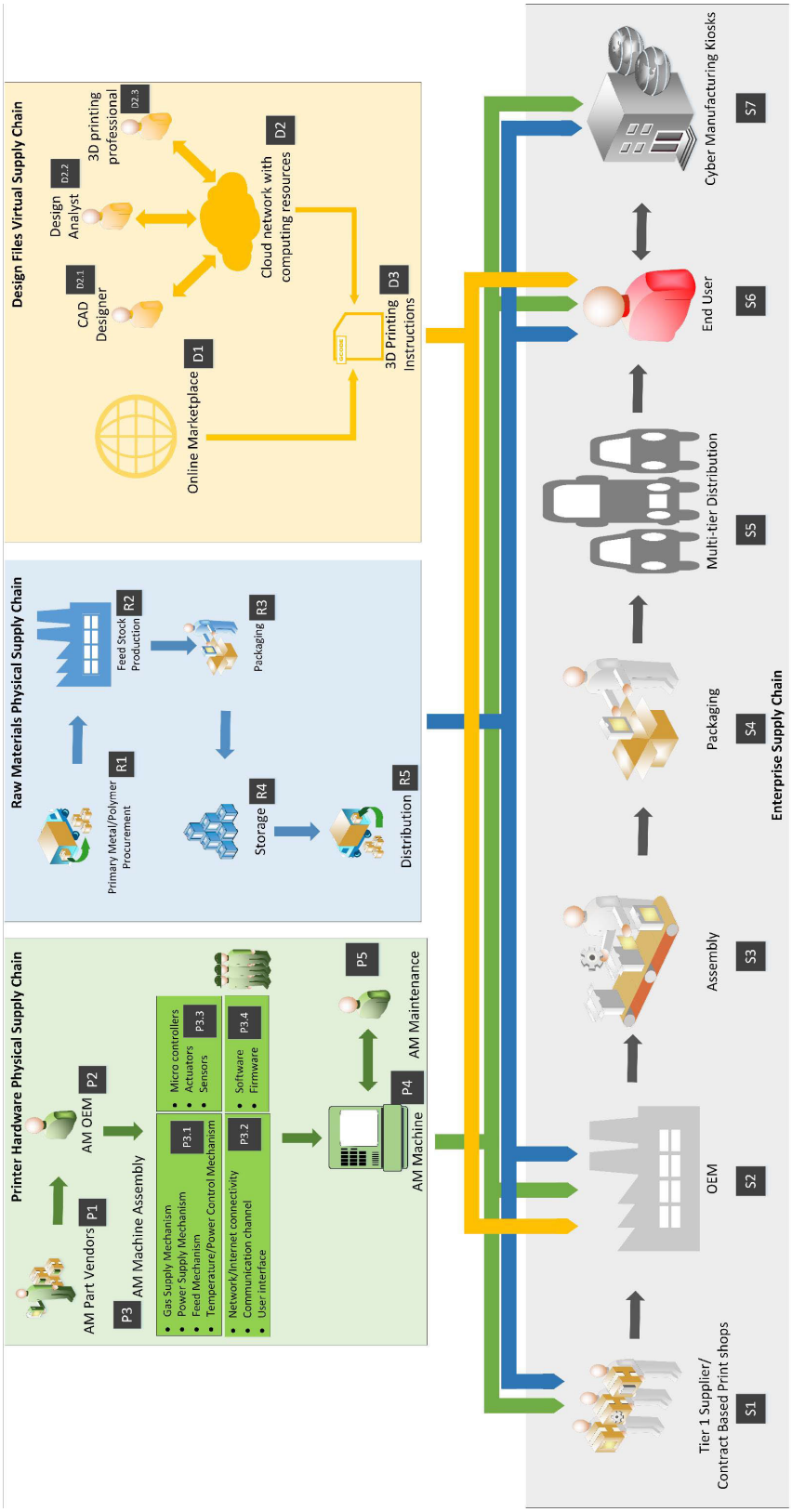

FIGURE 7. Holistic view of the extended AM supply chain.

of the produced part. Hence, an enhanced risk classification is presented in Fig. 8 to cover the attack classes for various supply chain components, along with featured examples of attacks.

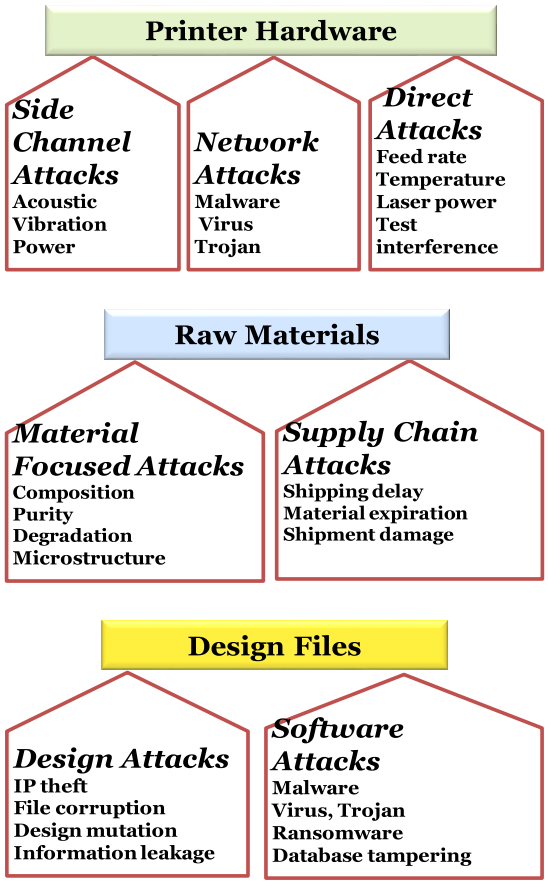

FIGURE 8. Type of attacks for each of the three supply chain components of AM.

1) PRINTER LEVEL ATTACKS

Some of the printer level attacks that are specific to AM include side channel attacks, where the attacker does not even need to access the printer and only uses the peripheral systems to conduct the attack. An example of such attack may include hacking a smart power meter that is located on the power line and belongs to the utility company. The power usage can reveal the type of component that is printed, decode the manufacturing schedule, and the manufacturing production run. Sabotaging the power supply can lead to defects in the components, production shutdown or damage to the printer. Other side channel attacks can reverse engineer the part from the acoustic signatures during 3D printing of the part [29]. P3.1, P3.3 and P4 are the vulnerable links of the supply chain for such attacks.

Since 3D printers are often connected to the internet for remote monitoring of print jobs, managing the print queue, and to diagnose problems, this is another vector that an attacker can exploit (P3.2). The attacker can plant malware, Trojans and viruses in the printer software or firmware (P3.4). The attacker can snoop and obtain confidential information on the product geometry, quality, production run, and manufacturing condition. The cyber link (P3.2) can be used to introduce defects in the components. One of the possibilities is to program the viruses to embed defects only in a small number of parts in a large production run. Traditionally, sampling methods are used to select a small number of parts from a large production run for testing. A careful programming of the virus to embed defects in a few parts can reduce the possibility of selection of one of these parts as the sample for testing. Failure of a few parts in a large batch can lead to recall of the entire batch and significant monetary losses for the company.

Networked printers can also be subjected to direct hacking attacks where raw material feed rate, extruder temperature, laser power, and other manufacturing conditions can be changed on the go to embed a defect in the part. In large size part with internal details, sometimes detection of internal defects may be very challenging due to the limitations of imaging and testing methods [30]. Such attacks may typically occur at S1, S2, S6 and S7 where the printer (P4) is procured and connected to an established internal network. The attackers take advantage of the vulnerabilities of P3.2.

2) RAW MATERIAL ATTACKS

The supply chain of raw material can be affected by attacks in two distinct ways. In the first type of attacks, logistics of supply chain are attacked (R1, R5) to cause problems such as shipment delays that can be detrimental to just-in-time manufacturing concept adopted by a number of industry sectors. There can also be other consequences of shipment delays such as expiration of raw material and imminent damage to packaging such as loss of temperature, moisture barrier and pressure conditions.

The second type of attacks are where the intent is to directly compromise the quality of raw materials. For example, attacks that can increase the oxygen content by hacking into the electrical system to change the oxygen valve settings, leading to surface oxidation of feed metal powder particles. Such an attack can severely degrade the raw material quality and happens at R2. Other attacks in this category can include switching the original high-quality feed material with counterfeit low-quality material or mixing impurities with the feed material. In metallic particles, microstructure of the particles is an important parameter, which can be affected to change the properties of the powder without causing any noticeable difference in the size distribution or appearance of the powder particles.

3) DESIGN LEVEL ATTACKS

Fig. 1 shows that a number of steps are conducted using software programs during development of the CAD solid model, which includes virtual design optimization (either based on optimization principles and algorithms or based on computational analysis), slicing and G-code generation. Any of these stages can be attacked to mutate the designs or steal the files for unauthorized production. While the original files can be used to produce high quality parts, the sabotaged files left in the cloud drive may result in low quality production of part by the OEM. These attacks happen at D2 and affect outputs from S1, S2, S6 and S7.

B. CLASSIFICATION OF RISK TYPES

The threat taxonomy presented in the previous section does not cover a number of risks involved in the AM sector. Since AM is a hardware sector, where end product is a hardware component, there are additional risks involved where a component can be legally obtained but used for reverse engineering to conduct counterfeit or unauthorized production of part of the same design. Here the end user (S6) is the vulnerability. These possibilities that are outside of the AM threats and attacks are included to expand Fig. 8 in developing a more comprehensive attack scenario in Fig. 9. Four threat classes are presented in Fig. 9 based on the interaction of attacker with the AM supply chain components.

The four threat categories identified for the AM CPS that cover the entire supply chain are:

- ***Side channel attacks***: The AM supply chain is not compromised at any level. The information is obtained from peripheral systems such as power meter, security camera, cell phone microphone, and vibration sensor [31]. These attacks may be targeted to steal the design by recreating the information from obtained side channel information or embed defects in the part during manufacture. Most of the attacks in this category require skills of hacking into the systems that have some level of embedded security such as power meter or cell phones. The AM Company may not be able to detect such attacks on third party. The location of such attacks are S1 and S2, but the means of conducting such an attack by the attacker may be ubiquitous.
- ***Direct sabotage***: This kind of attack can be conducted anywhere in the supply chain including at the printer during printing of part to introduce defects or in the cloud storage to conduct a design mutation (D2). Raw material supply can be attacked to degrade the material quality. These attacks require skills for hacking into computer systems and networks. The company may be able to deploy tools for defense against such attacks or take precautionary measures. Raw material attacks by remotely changing temperature setting of storage room (R4) or disabling moisture control unit may be examples of this category.

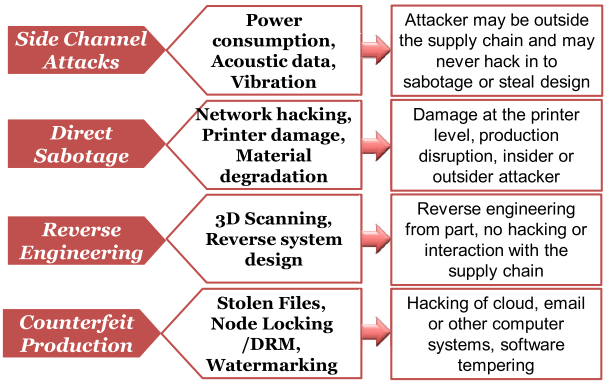

FIGURE 9. Classification of risk types in AM supply chain.

- ***Reverse engineering***: In this case, the attacker (S6) has no interaction with the supply chain except for legally acquiring the part. The part can be subjected to reverse engineering methods such as 3D scanning, computed-tomography (CT) scanning, or recreating CAD model from dimensional measurements. The CAD models thus developed can be used to produce replicas of the part. In this case, the company may not even know about the attacks for a long time. The skill set required for this kind of attack will require proficiency in CAD modeling and various imaging techniques that can be used for model reconstruction. The attacker needs to have facilities such as CAD design tools and imaging hardware. Developing product authentication methods is important for defense against this attack class.
- ***Counterfeit/unauthorized production***: A vulnerability specific to CPS is that the stolen digital files can be used to produce high quality parts that are exactly of the same quality as the original part. The quality of these products is not subjected to the skill of technician. The files can be stolen by hacking the cloud storage drive (D2) or intercepting email communication, among other possibilities. Even the genuinely contracted job (S1) shops may produce more than the contracted number of parts and sell in the gray market. Methods for developing secure and trustworthy CAD files is required for defense against this category. From solutions such as password protection and watermarking to more design based solutions that embed design features as security features are among the possible defense mechanisms. Although this is mostly a cyber-defense problem, this attack class can also be addressed by developing solutions where design features also act as security features against counterfeit production.

We have elucidated that there is a vulnerability associated with almost each of the links in the supply chains (P, R, D and S). Attacking just one of these vulnerabilities will lead to one or many of the risks presented in Fig. 9. Several illustrative examples presented in Fig. 9 show that encryption of files, node locking, digital rights management (DRM), watermarking, password protection and other file level security methods can make the supply chain secure for files but reverse engineering can be completely outside of control of the manufacturer. Network level security methods are also not able to address some of these threats. Hence, it is clear that the use of only traditional cybersecurity methods related to network security, access control and file protection is not sufficient for AM and a combination of multiple methods is necessary. These methods should continue to evolve over time to keep pace with the threats and emergence of methods to breach them.

C. SECURITY SOLUTIONS FOR AM

Here we discuss existing security solutions which the stakeholders at the respective stages of the supply chain can adopt to mitigate risk against some of the vulnerabilities.

1) ROBUST INFORMATION INFRASTRUCTURE

Knowledge of provenance of the materials in any supply chain is essential to qualify as a secure supply chain. Such knowledge must include a trace of transit of materials from origin to destination along with their state in each stage of transit. The metal AM industry relies on multiple supply and process chain to enable its operations (Fig. 7 – P, R, D and S) and therefore, the lack of this knowledge poses risks of counterfeiting and sabotaging of the intermediate raw material (R1, R5) or hardware procured by the businesses involved (P1). Tracing back of product for identifying failure point for future risk mitigation also becomes impossible.

Besides the establishment of the knowledge of provenance, there must be a consensus established across the various businesses and entities (P1, P2, P5, R1, R2, R5, D, and S1 through S6) involved to agree upon the provenance. The AM industry calls businesses from powder metallurgy/polymer, industrial design and electronics, each with their own unique process and supply chain. Convincing such diverse businesses to agree upon the accepted provenance and therefore identifying responsible businesses for a failed final 3D printed product becomes a challenge. There must be a trust established among these diverse businesses in the knowledge of provenance accrued and established for each 3D printed product.

The aforementioned issues, call for an information database, which holds a historical record of individual 3D printed products with details pertaining to the inputs consumed during its production and the process chain and the supply chain it went through. Such a system must also be able to bring together the distributed businesses on one platform which is robust against tampering attacks. The Blockchain technology can help to deliver on the issues identified. The Blockchain technology keeps track of transfer of ownership along with information relevant to the transfer. The Blockchain infrastructure is designed to be robust against tampering attacks and can successfully bring distributed entities to consensus. Smart contracts integrated with blockchain can further help in automatic execution of actions with secure authentication. This can bring the benefits of blockchain in process chains of manufacturers for secure communication among autonomous entities [32].

The architecture of this technology can also adapt to the requirements of the specific AM industry and its supply chain. It can be altered with respect to the consensus system, transaction capabilities, security & privacy, extensibility and identity management [33]. This flexibility of the technology can help in catering to the requirements of the multiple supply chain models discussed in the previous section.

2) ANTI-REVERSE ENGINEERING METHODS

AM produces hardware components, which can be purchased from the market and then be subjected to reverse engineering (at S6) for production of counterfeit parts. All available methods such as use of serial numbers and tracking codes can be implemented in AM field also. However, sometimes counterfeits are used in markets such as satellite components and medical implants, where company may not have access to the component to check authenticity once it is deployed. In addition, surface markings such as serial numbers and codes can be removed, altered, or duplicated. There have also been efforts reported that work with specialty materials designed to deter 3D scanning of the part. However, methods that rely on changing the material compositions or expensive specialty materials may not be the acceptable solutions in industrial products. Even a small quantity of nanoparticles can act as nucleation sites in metallic products and change the grain structure, which affects the mechanical properties and heat treatment regime.

Unique opportunities are available to encode information inside the product during layer by layer manufacturing. Although studies have shown that Quick Response (QR) codes can be embedded inside a part during manufacture [18], [34], large size of these codes is a concern to some people, especially if the size of the part is comparable to the size of the code itself. However, the embedded codes need not be as extensive as QR codes. Companies can design much smaller codes and embed them inside the part during manufacture, to read by methods such as radiography, ultrasound or CT scan. It is also noted that these codes need not be designed by creating voids in the product. In general, any recognizable signature can be used for such purpose, for example, structure and orientation of precipitates in a small localized area or a set of predefined areas in a metallic specimen, can be altered through printer settings and can be used for identification. Increasing interest in machine learning may lead to the possibility of using natural features of a given microstructure for product authentication purposes. Such methods are not yet fully developed but are increasingly being explored in the research community as the machine learning methods evolve and the AM process control improves.

3) SECURE AND TRUSTWORTHY CAD FILES

Protecting CAD files against unauthorized changes or sabotage is an important area. Often collaborating teams are sharing CAD files on cloud accounts (D2) for design work. In addition, customers are provided access to download files for printing (D1). At all times having trust in the integrity of the files is important. Methods are now available that can embed obfuscated design features in the CAD files to make them secure [35]. These secure files cannot be printed in high quality parts unless the security key is available. The features developed at the design level can be embedded at any step of the process such as in CAD files, in STL files, or in the G-code depending on the threat assessment for that supply chain. Similar authentication step will have to be developed to implement at an appropriate step. For example, a software utility implemented on the printer can authenticate the part for design mutations before printing. The security features can be in the form of one or more of the following: hidden

surfaces, direction of the surface vector in a STL file, angle of slicing with respect to x, y and z directions and slice thickness, among others. As the CAD files passes through various steps of process defined in Fig. 1, there is loss of information in the slicing and tool path creating processes. This possible loss of information can be used to develop features that may or may not survive at 3D printing stage and work for security. Securing the cloud (D2) and communication channels (P3.2) can be conducted by the cybersecurity professionals. The secure design strategy (implemented by D2.1) can provide additional security against printing of high-quality duplicate parts if the files are stolen by skilled hackers.

V. SUMMARY

Rapid advancements in the AM methods have contributed to changing supply chain scenarios for a large number of industrial products. Any supply chain is vulnerable to attacks that can lead to intellectual property theft, sabotage, counterfeit production and other threats. The present work is focused on analyzing the AM supply chain and identifying differences with respect to traditional manufacturing chains with the aim of understanding and classifying the threats. A supply chain model is proposed in this work where the supply chain for AM is divided into three streams: raw materials, 3D printer, and design files and a detailed analysis of these streams is provided at the granular level. The embedded IP can make the virtual supply chain more valuable than the physical supply chains. The detailed analysis on the holistic AM supply chain presents the interdependencies of the processes in the supply chain. The holistic model also illustrates how the three novel supply chain models. Businesses must decide where the digital version of the part is converted to the physical version by choosing appropriate supply chain models for their business. Depending on the resources and security infrastructure to mitigate risks, they can choose the varying lengths of the digital and the physical trail for the parts across the virtual supply chain and the physical supply chain respectively. Placing manufacturing closer to the customer resolves the physical supply chain risk to a great extent. On the other hand, manufacturing happening in the upstream alleviates much of the cyber threats associated with the virtual supply chain. In addition, the work is also focused on presenting a classification of threats across the entire AM supply chain into four categories. These categories are based on the level of interaction of the attacker with the AM supply chain and the skill set involved in implementing the attack and defending against it. This analysis is expected to help in developing solutions for various products. Several threat vectors and solutions are rooted in the design or product quality, which require involvement of mechanical engineers and materials scientists in developing defense methodologies. In cyber-physical systems such as AM, the cybersecurity threats can lead to hardware problems. Hence, the defense schemes also require a collaborative approach where security and design professionals need to work together.

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NIKHIL GUPTA received the Ph.D. degree in engineering science from Louisiana State University, specializing in lightweight advanced composite materials. He is currently a Professor with the Department of Mechanical and Aerospace Engineering, New York University Tandon School of Engineering. He is also affiliated with the Center for Cybersecurity and the Department of Civil and Urban Engineering. He has four issued and six pending patents. He has published more than 185 journal articles and book chapters. His current research projects are focused on cybersecurity in additive manufacturing and additive manufacturing security education and use of machine learning methods in materials characterization. As a materials scientist, he has been interested in developing lightweight advanced composites of metals and polymers for dynamic loading conditions. His research has been supported by the National Science Foundation, the Office of Naval Research, the Army Research Laboratory, and industry. He has served as a Membership Secretary of the American Society for Composites and the Chair of the TMS Composite Materials Committee.

AKASH TIWARI received the B.Tech. degree in industrial and systems engineering from the Indian Institute of Technology, Kharagpur, India, in 2019. He is currently pursuing the master's degree with the Department of Industrial and Systems Engineering, Texas A&M University, College Station, TX, USA.

He was a Summer Intern with the Royal Enfield Motors Factory, Chennai, India, in 2017. In 2018, he was a Summer Research Intern with the Durham Univeristy Business School, Durham, U.K.

SATISH T. S. BUKKAPATNAM received the bachelor's degree from S. V. University, Tirupati, India, and the master's and Ph.D. degrees from Pennsylvania State University, State College, PA, USA.

He has served as an AT&T Professor with Oklahoma State University and as an Assistant Professor with the University of Southern California. He is currently the Director of Texas A&M Engineering Experimentation Station (TEES) the Institute for Manufacturing Systems. He also holds an affiliate faculty appointment at Ecole Nationale Superior Arts et Metier (ENSAM), France. He also serves as a Rockwell International Professor with the Department of Industrial and the Systems Engineering Department, Texas A&M University, College Station, TX, USA. His research addresses the harnessing of high-resolution nonlinear dynamic information, especially from wireless MEMS sensors, to improve the monitoring and prognostics, mainly of ultraprecision and nanomanufacturing processes and machines, and cardiorespiratory processes. His research has led to 151 peer-reviewed publications (87 published/accepted in journals and 64 in conference proceedings), five pending patents, 14 completed Ph.D. dissertations, $5 million in grants as PI/Co-PI from the National Science Foundation, the U.S. Department of Defense, and the private sector, and 17 best-paper/poster recognitions. He is a Fellow of the Institute for Industrial and Systems Engineers (IISE) and the Society of Manufacturing Engineers (SME). He has been recognized with Oklahoma State University regents distinguished research, Halliburton outstanding college of engineering faculty, IISE Boeing technical innovation, IISE Eldin outstanding young industrial engineer, and SME Dougherty outstanding young manufacturing engineer awards. He also serves as an Editor of the *IISE Transactions*, Design and Manufacturing Focused Issue.

RAMESH KARRI (Fellow, IEEE) received the B.E. degree in ECE from Andhra University, and the Ph.D. degree in computer science and engineering from the University of California at San Diego. He is currently a Professor of electrical and computer engineering with New York University. He also co-directs the NYU Center for Cyber Security. He also leads the Cyber Security thrust of the NY State Center for Advanced Telecommunications Technologies, NYU. He co-founded the Trust-Hub and organizes the Embedded Systems Challenge, the annual red team blue team event. He has published more than 200 articles in leading journals and conference proceedings. His research and education activities in hardware cybersecurity include trustworthy ICs, processors and cyber-physical systems, security-aware computer-aided design, test, verification, validation and reliability, nano meets security, hardware security competitions, benchmarks and metrics, biochip security, and additive manufacturing security.